

Cardiac Field Artifact Removal in Ear-EEG Using Repetitive Disturbance Rejection

Wenlin Zhang¹, Samuel Lin^{1,6}, Zheng Wang^{1,2}, Shuai Li³, Ye Yuan¹,
Ziteng Ni⁴, Heng Kang Yao^{2,4}, Yongcheng Li⁵, Davide Momi⁷

Abstract—Continuous wearable EEG is increasingly paired with consumer wearables that provide a heartbeat stream (e.g., smartwatch ECG/PPG). Given the available heartbeat timing reference, we propose a repetitive-control framework that models Cardiac Field Artifact (CFA) as a heartbeat-aligned periodic disturbance, and we adopt iterative learning control (ILC) to clean up the artifact in a streaming fashion. After the mapping of measured EEG samples to the corresponding normalized cardiac phase around ECG R-peaks, phase-synchronous averaging then preserves the heartbeat-correlated component as a model of repetitive CFA disturbances. We propose an ILC based approach to track the averaged repetitive disturbance exponentially which enables the real-time CFA cancellation from EEG streaming. The effectiveness of proposed ILC-based CFA removal is demonstrated by the earEEG dataset using only the locked R-peak timing without the synchronized ECG waveforms.

I. INTRODUCTION

Wearable EEG is moving toward continuous, day-scale monitoring, and many users already wear a second device that provides heartbeat timing from ECG or PPG. This creates a practical path for low-channel ear-EEG denoising: use heartbeat timing already available from the wearable ecosystem rather than requiring additional EEG channels. In ear-centered montages, cardiac field artifact (CFA) is often prominent because electrodes are close to neck and vascular pathways. As one of the primary contaminations in wearable ear-EEG, CFA is the volume-conducted electrical field of the heart [1], distinct from the scanner-related ballistocardiogram artifact emphasized in EEG-fMRI work [2].

Classical ECG-referenced adaptive cancellation is well established [4]–[6], and it is a natural baseline. In our target regime it faces three practical limits: earables are channel-starved for blind source separation [7], learned denoisers are benchmarked mainly on high-density montages [8], and—most restrictive for a wearable deployment—reference-driven cancellation must be driven by the ECG *waveform*, sample-aligned to the EEG. Beat timing, by contrast, is a single scalar per cardiac cycle that crosses the device boundary intact, and the artifact remains phase-locked to it. This motivates a timing-only control view: model cardiac contamination as a repetitive disturbance and reject it by heartbeat phase indexing.

Affiliations: ¹Institute of Brain and Mind, CA, USA; ²Krembil Centre for Neuroinformatics, Centre for Addiction and Mental Health, Toronto, Ontario, Canada; ³University of Oulu, Oulu, Finland; ⁴Synchroni Scientific Instruments Inc., Canada; ⁵Shenzhen Institutes of Advanced Technology, China; ⁶Palos Verdes Peninsula High School, CA, USA; ⁷University of California, Irvine, CA, USA; Corresponding author: Wenlin Zhang (e-mail: william@ibmind.org).

Repetitive control and iterative learning control (ILC) are mature tools for periodic disturbance rejection in mechatronics [9], and they map naturally to the heartbeat cycle when reliable beat timing is available. In this work, we focus on phase-aligned template learning/cancellation driven by heartbeat timing in real time streaming EEG scenarios, especially in the low channel setup. The resulting contribution is a compact, wearable-oriented CFA removal scheme aligned with realistic device constraints: low channel count, low compute budget, and timing references that may come from a separate wearable.

II. PROBLEM FORMULATION

Following repetitive-control modeling conventions [9] [10], we model the measured ear-EEG signal as

$$y(t) = s(t) + d_r(t) + n(t), \quad (1)$$

where $s(t)$ is the neural signal of interest, $d_r(t)$ is the heartbeat-synchronous repetitive cardiac disturbance, and $n(t)$ aggregates asynchronous interference and sensor noise. Explicit generative models of the ECG waveform itself [11] are deliberately not required here. The subscript r denotes the repetitive nature of the disturbance. Its period is heartbeat dependent and is set by ECG R-peak timing, with beat interval

$$T_k = t_{k+1} - t_k, \quad (2)$$

where t_k is the k -th detected R-peak instant.

Assumption 1. *In ear-EEG, CFA is dominated by the repetitive component $d_r(t)$. The R-peak phase-asynchronous disturbances and noise $n(t)$ are assumed zero-mean under the R-peak-synchronous averaging, and DC offset and slow drift are removed by preprocessing.*

Using heartbeat phase, we represent the repetitive disturbance as a phase-indexed waveform

$$d_r(t) = d_r(\phi(t)), \quad \phi(t) = \frac{t - t_k}{T_k}, \quad t \in [t_k, t_{k+1}), \quad (3)$$

with $\phi(t) \in [0, 1)$ the R-peak-aligned cardiac phase.

Assumption 2. *We assume the neural signal can be decomposed as $s(t) = s_r(t) + s_n(t)$, where $s_r(t)$ is the heartbeat-repetitive neural component and $s_n(t)$ is the non-repetitive neural component of interest. The targeted applications in this paper do not seek to recover $s_r(t)$, so it is dropped from the problem formulation, then we obtain $\frac{1}{N_r} \sum_{k=1}^{N_r} s_k(\phi) = 0$.*

Assumption 3. The beat period T_k is assumed to be accurately locked, and slowly varying due to respiratory sinus arrhythmia (RSA), with additional small modulation from low-level muscle/motion effects in resting-state recordings.

Problem 1. Find the repetitive cardiac artifact template $\hat{d}_r(\phi(t))$ such that subtracting it from the measured signal recovers the non-repetitive neural signal of interest, after dropping the out-of-scope repetitive neural component $s_r(t)$:

$$\hat{s}_n(t) = y(t) - \hat{d}_r(\phi(t)) \approx s_n(t). \quad (4)$$

Under Assumptions 1–3, $\hat{d}_r(\phi(t))$ can be estimated from the R-peak-locked structure of $y(t)$ alone.

Note 1. Since $d_r(\phi(t))$ is periodic in the cardiac phase $\phi(t)$ indexed solely by R-peak instants $\{t_k\}$, Problem 1 depends only on R-peak timing and never on the ECG waveform shape. Consequently, any wearable heartbeat source—smartwatch ECG, wrist PPG, or any device delivering beat timing via a synchronization mechanism—is sufficient to instantiate the solution.

Note 2. In reality, the out-of-scope $s_r(t)$ may contain a phase-synchronous component, e.g. the heartbeat-evoked potential (HEP)—a genuine cortical response time-locked to the cardiac cycle [12]. This component may be absorbed into $\hat{d}_r$; preserving it is a meaningful extension worthy of further investigation.

To instantiate Problem 1, let $\hat{t}_k$ be the estimated k -th R-peak instant, $\hat{T}_k = \hat{t}_k - t_{k-1}$. Retaining the beat duration T_k , we map the k -th beat from time relative to its R-peak into normalized phase, with the R-peak placed at the center of the phase support rather than at its edge, set $\phi \in [-c, 1-c]$, $c \in (0, 1)$, we obtain

$$d_r^{(k)}(\phi) \triangleq \begin{cases} d_r(\hat{t}_k + \phi \hat{T}_k), & \phi \in [-c, 0), \\ d_r(t_k + \phi T_k), & \phi \in [0, 1-c), \end{cases} \quad (5)$$

Note 3. For strict real-time scenarios, Eq. (5) is non-causal before the t_k arrival, therefore the $\hat{t}_k, \hat{T}_k$ and $\hat{\phi}$ have to be predicted. If the application allows one heart beat delay that Eq. (5) can be applied causally with arrived and locked t_k beats and no estimated notation is needed. A forward-weighted split such as $\phi \in [-0.3, 0.7]$ as $c = 0.3$ might be more suitable for capturing the asymmetry of the cardiac cycle. We treat c as an implementation parameter.

Note 4. For offline non-causal processing, assuming R-peaks are accurately detected and bad beats are rejected, we average beats on a common normalized phase grid across N_r beats:

$$\hat{d}_r(\phi) = \frac{1}{N_r} \sum_{k=1}^{N_r} y_k(\phi), \quad y_k(\phi) \equiv y(\hat{t}_k + \phi T_k). \quad (6)$$

Substituting the signal $y_k(\phi)$ by its definition from the discrete time version of Eq. (1), we obtain $\hat{d}_r(\phi) \equiv d_r(\phi)$, as $s_{n,k}(\phi)$ and $n_k(\phi)$ are averaged to zero over time. For more detailed analysis, we can refer to the established template-based BCG/CFA artifact-removal idea of [3], and we will focus on real-time solution of Problem 1 in the next sections.

III. REPETITIVE DISTURBANCE REJECTION

A. Buffered Template Averaging

For streaming template learning, the batch average of Eq. (6) is replaced by its standard exponentially-weighted ILC recursion, updated once per learnable beat at each phase ϕ :

$$\hat{d}_r^{(k)}(\phi) = \hat{d}_r^{(k-1)}(\phi) + a [y_k(\phi) - \hat{d}_r^{(k-1)}(\phi)], \quad (7)$$

where $a \in (0, 1]$ is a tunable learning gain. The supervisor freezes the update on premature/missed beats.

Note 5. In streaming operation, the averaging in Eq. (7) has to run with one beat buffered, as in strict real-time the k R-peak has not arrived yet. Once the averaging converges, this ILC recursion converges to the exponentially averaged normalized-phase template of Ref. [3].

B. Real-Time Cancellation

For applications where one beat of delay is acceptable, buffered cancellation is a strong candidate: when the current R-peak arrives, the preceding learnable beat is cancelled using the template available before that beat, and is then used to update the template for the following beat. The completed beat is anchored to its detected R-peak, so there is no anchor-prediction error; its cleaned output is emitted with one heart-beat of latency:

$$\hat{s}_k(\phi) \approx y_k(\phi) - \hat{d}_r^{(k)}(\phi). \quad (8)$$

The equation is timestamp-aligned for comparison: in a streaming implementation, the cleaned samples of beat k are emitted from a buffer only when t_{k+1} arrives. The buffered mode is therefore preferred whenever the application can tolerate one-beat delay.

C. Strict real-time cancellation.

When zero added latency is required, we set up the strict real-time cancellation at current t , anchored on the the next incoming R-peak at t_{k+1} , we obtain

$$\hat{s}(t) = y(t) - \hat{d}_r^{(k)}(\hat{\phi}^{(k+1)}(t)). \quad (9)$$

Note 6. In Eq. (9), we only need to run predicted cancellation for $t \in [\hat{t}_{k+1} - c\hat{T}_{k+1}, \hat{t}_{k+1})$ before the t_{k+1} arrives, when the t_{k+1} arrives, it can run the cancellation based on converged $\hat{d}_r^{(k)}$, which we denote the k -th $\hat{d}_r$ as one beat delayed template. $\hat{d}$ includes a phase de-normalization to be subtracted from the signal $y(t)$, we omit the notation for simplicity.

For predicting $\hat{\phi}(t)$ for $t \in [t_{k+1} - cT_k, t_{k+1})$, we define the observed interval z_k as $z_k = \hat{t}_k - \hat{t}_{k-1}$, and we assume the measurement model as

$$T_k = T_{k-1} + w_k, \quad w_k \sim (0, q), \quad (10a)$$

$$z_k = T_k + v_k, \quad v_k \sim (0, r). \quad (10b)$$

Here q is the beat-to-beat RR-increment variance, and r is the R-peak timing-jitter variance. We set up the PI tracker as:

$$\frac{d\hat{\phi}}{dt} = \frac{1}{\hat{T}_{k-1}}, \quad (11a)$$

$$\hat{\phi} \leftarrow \hat{\phi} - K_p e_{\phi,k}, \quad (11b)$$

$$e_{\phi,k} = \frac{z_k - \hat{T}_{k-1}}{\hat{T}_{k-1}}, \quad (11c)$$

$$\hat{T}_k \leftarrow \hat{T}_{k-1} + K_{ss} (z_k - \hat{T}_{k-1}). \quad (11d)$$

Note 7. Eq. (11) provides a basic framework for predicting the $\phi(t)$ for the $k+1$ peak cancellation based PI tracking before the arrival of the t_{k+1} peak. Eq. (11d) is a first order tracking of $\hat{T}_k$ and the gain K_{ss} can be obtained by scalar Kalman filter with pre-calibrated q and r as leaky I control, and k_p applied in Eq. (11b) acts as the P control on the actual phase index. Higher order tracking algorithm could be designed for certain application needs, and when k_p and K_{ss} set to zero, it will fall back to re-use previous R-peak anchored phase.

Note 8. To prevent abnormal beats from corrupting the timing estimate or the repetitive template, we use the normalized RR innovation as a gating metric:

$$\nu_k = \frac{z_k - \hat{T}_k}{\sqrt{P_k^- + r}}. \quad (12)$$

Here P_k^- is the prior RR-estimation covariance maintained by the scalar Kalman implementation. When $|\nu_k|$ exceeds a threshold η , the beat is treated as unreliable: the R-peak predictor and template update may be frozen, or cancellation could fall back to re-use previous R-peak anchored phase indexes.

Note 9. The streaming real-time implementation of the iterative cancellation lookup and subtraction require $O(1)$ work per sample and $O(n_t)$ memory. The scalar Kalman PI predictor will be offline calibrated or have dedicated online calibration sessions together with possible high order IIR filters for tracking will keep the calculation cycle manageable for embedded applications.

IV. EXPERIMENTS

To evaluate the method we implemented a robust R-peak detector and PI phase tracker for both real and simulated data [13]. We will showcase the effectiveness of the proposed real-time repetitive CFA cancellation method in this section.

A. Real Ear-EEG Validation

In this subsection, we use the two-channel ear-EEG configuration of the SeizeIT2 dataset, a multicenter wearable epilepsy corpus of long-duration multimodal recordings (behind-the-ear EEG, ECG, EMG, ACC, GYR) with focal-seizure annotations across a large cohort [14]. Each EEG channel is high-pass filtered at 0.5 Hz, notch filtered at 50 Hz and resampled onto the ECG sample grid, and only supervisor-approved learnable beats enter the average.

Buffered and Strict Real-Time CFA Cancellation (SeizeIT2 ds005873)

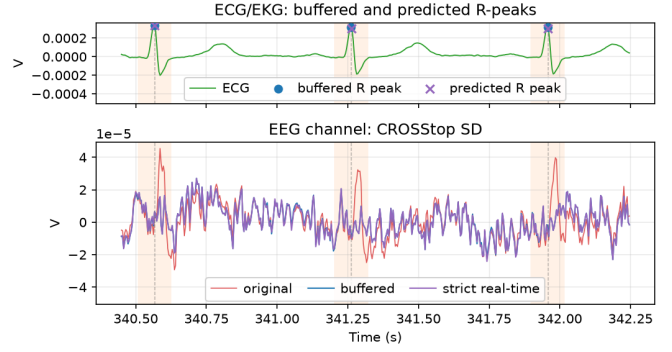


Fig. 1. Buffered and strict real-time repetitive CFA cancellation on a short SeizeIT2 segment (ds005873, sub-070, ses-01, run-20, behind-the-ear channel; 340.45–342.25 s). Top: chest ECG (green) with buffered detected R-peaks (blue circles) and predicted strict-real-time R-peaks (purple crosses). Bottom: original (red) and buffered (blue) and strict real-time (purple) ear-EEG. Both outputs use the repetitive template of Eq. (7).

Fig. 1 shows buffered and strict-real-time repetitive disturbance rejection on the SeizeIT2 segment. The template is learned from $N_r = 851$ learnable beats over a 600 s window; the strict-real-time trace uses the predicted timing anchor and therefore exposes the timing-jitter penalty relative to the buffered trace.

Template SNR via a phase-scramble null. Since the true disturbance d_r is unobservable on real recordings, we validate the template against a null that shares its statistics but destroys R-phase locking. Define the R-locked energy of a signal y under a peak set P on the normalized-phase grid Φ ,

$$\mathcal{E}_{true}[y] = \sum_{\phi \in \Phi_{lobe}} \left[\frac{1}{|P|} \sum_{k \in P} y_k(\phi) \right]^2, \quad (13)$$

where Φ_{lobe} is the normalized phase grid corresponding to the main-lobe window and $y_k(\phi)$ includes the phase denormalization or interpolation needed.

For each scramble realization b , we circularly shift the complete peak train to obtain $P_{scr}^{(b)}$; this preserves the RR-interval sequence but destroys its absolute alignment with y , and we calculate $\mathcal{E}_{scr}^{(b)} = \mathcal{E}_{P_{scr}^{(b)}}[y]$. The reported null floor is $\overline{\mathcal{E}_{scr}} = B^{-1} \sum_{b=1}^B \mathcal{E}_{scr}^{(b)}$, with $B = 200$, we obtain

$$Q = \frac{\mathcal{E}_{true}}{\overline{\mathcal{E}_{scr}}}, \quad \text{SNR}_{tmpl} = 10 \log_{10} Q, \quad (14)$$

and a bias-corrected estimate of the CFA energy is $\max(\mathcal{E}_{true} - \overline{\mathcal{E}_{scr}}, 0)$. We further define a bias-corrected CFA RMS, a scale-free measure of the artifact the template removes:

$$A/SD = \frac{\sqrt{\max(\mathcal{E}_{true} - \overline{\mathcal{E}_{scr}}, 0)/|\Phi_{lobe}|}}{SD}. \quad (15)$$

The subtraction removes the leakage bias and $SD = \text{std}(y)$ is computed over the full preprocessed segment, so A/SD is the R-locked deflection in units of EEG standard deviation.

The denominator $|\Phi_{\text{lobe}}|$ is the number of phase-grid points in the lobe. For the real-data example in Fig. 1, the metric reads $\text{SNR}_{\text{templ}} = 28.80$ dB and $A/\text{SD} = 1.061$, indicating a strong and clearly resolvable R-locked component. Applying the same calculation to the one-beat-buffered and strict-real-time outputs gives -1.92 dB/0.000 and 19.11 dB/0.328, respectively. Across the listed cohort, we find 365 runs from 36 subjects exceeded 20 dB, with mean $\text{SNR}_{\text{templ}} = 22.67$ dB (median 22.19 dB) and $A/\text{SD} \in [0.110, 1.220]$ with mean $A/\text{SD} = 0.557$. These values were regenerated using the raw, uncanceled EEG and 200 phase-scramble realizations per run. Applying the same 200-scramble calculation after cancellation to all 365 runs gives a mean buffered residual SNR of -8.31 dB (median -8.29 dB), with no runs remaining above 20 dB; the corresponding buffered A/SD has mean 0.00061 (median 0.000), where the near-zero values reflect the nonnegative leakage-bias correction. For strict real-time cancellation, the mean SNR is 17.57 dB (median 18.20 dB), with 110 runs above 20 dB, and mean $A/\text{SD} = 0.339$ (median 0.324). The decrease from the raw values indicates removal of the R-locked component; the strict-mode residual is higher because its pre-R cancellation interval uses predicted rather than detected R-peak timing, the design of better R-peak predictor will be beneficial future work.

Coherence view. Fig. 2 contrasts the same R-locked ECG window against three EEG views: the raw ear-EEG R-locked raw waveform, the stacked non-causal template, and the proposed buffered one-beat-delay template. Please note that non-causal mode aligns with existing method in [3], and the buffered one-beat-delay template achieves exponentially averaged template correspondingly, as shown in the left subplot. The right subplot shows the coherence between the R-locked ECG window and the three EEG views, respectively. As coherence is the key enabling factor to the waveform referenced RLS type of adaptive cancellation. We could see the proposed buffered one-beat-delay template iteratively learned in real-time achieves comparable coherence level to the existing non-causal template, and the raw EEG window shows diluted coherence except the 20Hz centered lobe.

B. Synthetic timing validation

We now sweep the ECG-to-artifact channel from linear (high coherence) to strongly nonlinear (near-zero coherence) at fixed artifact power and compare the three timing-based cancellation modes. ICA-based CFA removal is not listed here, as it requires sufficiently many channels and is outside this low-channel-count setting. Fig. 3 compares RLS, non-causal, one-beat-buffered, and strict-real-time cancellation under controlled predictability. RLS uses the sample-aligned ECG waveform as a reference, whereas the three proposed timing-based modes estimate the heartbeat-locked disturbance directly from the EEG using beat timing. The proposed buffered cancellation achieves slightly worse but comparable performance to the non-causal cancellation [3]. Strict real-time cancellation apparently has room for improvements but

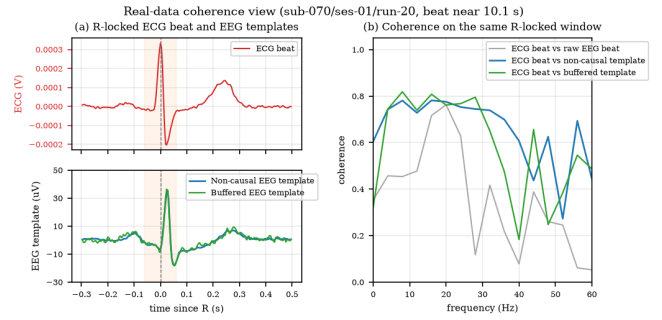


Fig. 2. Coherence view on the same real recording. Left: one R-locked ECG window overlaid with the non-causal EEG template and the buffered one-beat-delay EEG template. Right: coherence between the windowed ECG beat and three R-locked EEG views: the raw EEG beat locked window, the stacked non-causal template, and the buffered template. Both templates lock to the same R-centered structure and retain appreciable coherence with the ECG at lower frequencies, while the raw beat is diluted by the neural background outside the 20Hz centered lobe.

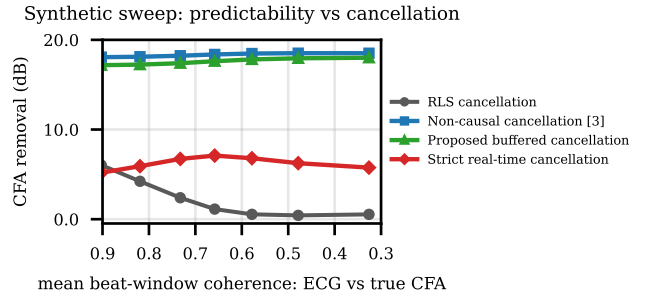


Fig. 3. Synthetic sweep of reference predictability versus cancellation performance. The sweep blends an ECG-coherent linear artifact with a phase-locked but waveform-decorrelated component at fixed artifact power; the x-axis is the mean 1–20 Hz coherence between the averaged ECG beat and the averaged *true* CFA beat, and the y-axis is CFA component removal in dB calculated from the synthetic simulation

also outperforms the RLS baseline which needs aligned ECG waveforms..

V. CONCLUSION

We have presented a control-theoretic view of cardiac-field-artifact (CFA) removal in low-channel ear-EEG: CFA is modeled as a heartbeat-phase-locked repetitive disturbance and rejected by phase-indexed synchronous averaging and iterative learning, anchored only to R-peak timing only without ECG waveform in streaming EEG setup. By improving SNR in the low-channel ear-EEG setting, it supports non-invasive wearable BCI applications such as brain-state monitoring and real-time low-channel whole-brain modeling [15], especially online neural feedback applications.

More broadly, this work aims to bridge mature mechatronic disturbance-rejection theory and biosignal denoising, opening a control-systems avenue for wearable neural interfaces for low channel EEG real-time cleanup.

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